Chapter 18, Problem 6RQ

Problem

Evaluating the integral $\int_{-\infty}^{\infty} \frac{10\delta(\omega)}{4+\omega^2} d\omega$ results in:

(a) 0

(b) 2

(c) 2.5

(d) ∞

Step-by-step solution

Step 1 of 1

Consider the following integral.

$$\int_{-\infty}^{\infty} \frac{10\delta(\omega)}{4+\omega^2} d\omega$$

The following is the sampling or sifting property of the impulse function.

$$\int_a^b f(t)\delta(t-t_0)\,dt = f(t_0)$$

Use the sampling or sifting property of impulse function to evaluate the integral.

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{10\delta(\omega)}{4+\omega^2} d\omega &= \int_{-\infty}^{\infty} \left(\frac{10}{4+\omega^2}\right)\delta(\omega-0)\,d\omega \\ &= \left(\frac{10}{4+\omega^2}\right)_{\omega=0} \\ &= \frac{10}{4} \\ &= 2.5\end{aligned}$$

Thus, the integral $\int_{-\infty}^{\infty} \frac{10\delta(\omega)}{4+\omega^2} d\omega$ gives $\boxed{2.5}$ and the correct choice is $\boxed{c}$.